WEEK 2 · LESSON 10 OF 24 · HANDOUT

Advanced Prompting: Examples, Steps and Iteration

Study time: **about 50 minutes** (reading, video and activity)

▶ **Watch:** https://www.youtube.com/watch?v=_ZvnD73m40o — open the lesson online to watch inside the course.

🎓 **Go deeper:** Harvard CS50 AI, Lecture 6: Language (attention and transformers) — https://www.youtube.com/watch?v=QAZc9xsQNjQ

Your goal in this lesson: Use advanced techniques — examples (few-shot), step-by-step reasoning, asking the AI to question you, critique-and-improve, and reusable templates.

Video note: the video is a full tutorial. Watch the first 20 minutes for this lesson; the rest is optional.

1. Show, don't just tell (few-shot prompting)

If you want a specific style, give one or two **examples**.

> *"Write product descriptions for our shop. Here is the style I want: Example: 'Imigongo wall art — handmade in Kirehe with natural pigments. Bold geometric patterns that bring Rwandan heritage to any room.' Now write descriptions in the same style for: a woven basket, a coffee mug, a kitenge tote bag."*

The AI copies the length, tone and structure of your example.

2. Ask for step-by-step thinking

For problems with several steps — calculations, plans, decisions — ask the AI to **work step by step** and show its reasoning. This reduces careless mistakes and lets you check each step.

> *"A student pays 45,000 RWF per month for rent and 30,000 RWF for food, and earns 120,000 RWF. Work step by step: what is left each month, and what percentage of income is spent on rent?"*

3. Let the AI interview you

When you are not sure what details to give, flip it around:

> *"I want to write a funding proposal for a youth coding club in Rubavu. Before you write anything, ask me the 5 questions you need answered to do this well."*

You answer the questions, then the AI writes a much better proposal.

4. Critique, then improve

Ask the AI to review its own work (or yours) against clear criteria:

> *"Review the proposal above as a strict funder. List its 3 biggest weaknesses, then rewrite it to fix them."*

5. Break big tasks into a chain

Instead of "write my whole report", work in stages:

1. "Suggest an outline for a report on…"
2. "Expand section 2 using these notes…"
3. "Write a 100-word summary of the full report."
4. "Check the report for unclear sentences."

Each step is easier to check than one giant answer.

6. Save reusable templates

When a prompt works well, save it with blanks you can fill in later:

> *"You are a [role]. Write a [document type] for [audience] about [topic]. Use a [tone] tone, keep it under [N] words, and format it as [format]."*

The iteration loop

Ask → Read → Check → Refine → Repeat. Great results rarely come from the first prompt. They come from two or three rounds of clear feedback, such as *"shorter"*, *"more formal"*, *"add an example from agriculture"*, *"turn this into a table"*.

Remember this: Examples, step-by-step reasoning, interviews, critique and chaining turn an average AI answer into professional work.

Name: _____ Date: _____

Activity

Task 1

Use technique 3: ask the AI to interview you (5 questions) before writing something real — a CV summary, a club announcement or a short proposal.

Task 2

Answer its questions and get the draft.

Task 3

Use technique 4: ask it to critique the draft as a strict reviewer and rewrite it.

Task 4

Turn the best prompt into a **template** with blanks and save it for your prompt library.

Self-check

Think about your answers here, then take the quiz online — it is marked instantly and explains every answer.

1. Giving the AI one or two examples of the style you want is called…
   - A. Reverse coding
   - B. Data mining
   - C. Zero effort prompting
   - D. Few-shot prompting
2. Why ask the AI to work 'step by step' on a calculation?
   - A. It makes the answer longer for fun
   - B. It reduces careless errors and lets you check each step
   - C. It is required by law
   - D. It makes the AI faster
3. What is the benefit of asking the AI to interview you first?
   - A. It avoids writing anything
   - B. It saves the AI's time
   - C. It gathers the details needed for a much better result
   - D. It makes the AI more creative only
4. Breaking a big report into outline, sections, summary and review steps is called…
   - A. Chaining tasks
   - B. Hallucinating
   - C. Supervised learning
   - D. Formatting
5. What is a prompt TEMPLATE?
   - A. A saved prompt with blanks you fill in for new tasks
   - B. An AI's training data
   - C. A type of image
   - D. A paid AI plan
6. Which sequence best describes the iteration loop?
   - A. Ask → Accept → Submit
   - B. Copy → Paste → Done
   - C. Check → Ask → Forget
   - D. Ask → Read → Check → Refine → Repeat

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